Physics-Informed Shape Optimization of Blade Paul Traps

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1. Abstract

Trapped-ion platforms remain one of the most mature and promising architectures for quantum technologies, including quantum computation, simulation and networking. Among the broad family of radio-frequency ion traps, blade Paul traps are especially attractive because they offer high optical access, and experimentally proven versatility for both single-ion and multi-ion operation. However blade Paul traps have weaker confinement than rod traps.

This proposal addresses that gap by introducing a fully differentiable framework for the *physics-informed shape optimization of blade Paul traps*. The central idea is to couple:

- an adaptive physics-informed solver based on **Physics-Informed Gaussians (PIGs)** for the electrostatic and pseudopotential fields, and
- a **Coordinate Projection Network (CPN)** for continuous shape deformation and optimization.

The motivation for this combination is strong. Classical PINNs are elegant but often suffer from spectral bias and can struggle with sharp gradients, high-frequency behavior, or localized singular structure. PIGs address this by replacing the purely MLP-centered representation with adaptive Gaussian feature embeddings whose means and variances move during training toward physically difficult regions. This makes them particularly well suited to blade traps, where strong field variation appears near electrode edges and tips. In parallel, coordinate projection offers an elegant way to parameterize shape deformation without introducing artificial material smoothing, while keeping geometry differentiable throughout optimization.

The outcome of the project seeks to be an open, modular implementation capable of:

1. solving the electrostatic field of blade Paul trap geometries with a PIG-based physics-informed solver,
2. optimizing blade geometry under trap-quality objectives using a CPN-based shape deformation module,
3. benchmarking against conventional geometries and validating results against numerical baselines.

In short, this project aims to turn trap design from a manual simulation loop into a differentiable optimization pipeline.

2. Paul Traps

2.1. Why Blade Paul Traps?

Blade Paul traps are a class of symmetric radio-frequency traps in which thin electrode blades generate a quadrupole-like field suitable for confining ions. Compared with rod traps, blade traps can offer improved optical access and tighter electrode spacing, which are highly desirable

in experiments involving laser access, imaging, and photon collection. They have been used in quantum simulation, precision measurement, and quantum information experiments, and segmented variants allow additional control over axial potentials and ion positioning.

From a design perspective, blade traps are compelling because their performance is highly geometry-dependent. Small changes in blade angle, tip radius, separation, segmentation, and relative alignment can alter the local field curvature, pseudopotential depth, micromotion sensitivity, and optical accessibility. This makes them a natural candidate for differentiable shape optimization.

2.2. Electrostatic Forward Model

We consider the vacuum electrostatic potential $\Phi(\mathbf{x})$ generated by electrodes in a domain $\Omega \subset \mathbb{R}^d$, with $d = 3$. In the vacuum region away from free charge, the governing equation is Laplace's equation:

$$\nabla^2 \Phi(\mathbf{x}) = 0, \qquad \mathbf{x} \in \Omega_{\text{vac}}. \tag{1}$$

Boundary conditions are prescribed on the electrode surfaces:

$$\Phi(\mathbf{x}) = V_{\text{RF}} \cos(\Omega_T t) \quad \text{on RF electrodes,} \tag{2}$$

$$\Phi(\mathbf{x}) = V_{\text{DC},k} \quad \text{on DC or segmented control electrodes,} \tag{3}$$

with appropriate outer-boundary truncation conditions, depending on the domain setup.

Since direct time-dependent simulation of the RF dynamics is typically not needed for geometry optimization, we solve for the *RF amplitude potential*

$$\Phi_{\text{RF}}(\mathbf{x}) \tag{4}$$

such that the time-harmonic physical potential is

$$\Psi(\mathbf{x}, t) = \Phi_{\text{RF}}(\mathbf{x}) \cos(\Omega_T t). \tag{5}$$

2.3. Pseudopotential Approximation

Near the trap center, the time-averaged ponderomotive pseudopotential for an ion of charge q and mass m is

$$U_{\text{ps}}(\mathbf{x}) = \frac{q^2}{4m\Omega_T^2} \|\nabla \Phi_{\text{RF}}(\mathbf{x})\|^2. \tag{6}$$

This quantity defines effective confinement in the radial directions. Good trap geometries are those for which the pseudopotential has:

- a well-defined and centered RF null,
- strong but controlled local curvature,
- acceptable depth and secular frequencies,

- robustness to small perturbations and misalignments.

2.4. Secular Motion and Stability Quantities

In an ideal quadrupole setting, ion motion can be described by the standard Mathieu equations. However, for geometry optimization, it is not necessary to simulate full ion trajectories. Instead, stable confinement can be characterized through field-derived quantities computed from the RF amplitude potential and the associated pseudopotential.

In this work, we focus on metrics that directly correspond to the optimization objectives used in the learning framework:

- **RF null position:** The location where the electric field vanishes, ensuring minimal micromotion and proper centering of the trap.
- **Pseudopotential depth:** The effective trapping depth defined by the pseudopotential barrier, which determines the robustness of confinement against escape.
- **Geometric accessibility constraints:** Measures related to electrode proximity and opening angles, which impact optical access and experimental usability.

These quantities provide a physically meaningful and computationally efficient surrogate for full dynamical simulation, and are directly used to define the optimization losses guiding the geometry design.

3. Technical Approach and Model Design

3.1. Overview of the Proposed Architecture

The proposed system consists of two tightly coupled modules:

1. a **PIG-based field solver** that predicts the RF amplitude potential and associated field quantities in the domain,
2. a **shape optimization module** based on a Coordinate Projection Network, which maps a reference blade geometry into an optimized geometry.

3.2. Physics-Informed Gaussians (PIGs) as the Field Solver

3.2.1. Motivation

PIGs are an attractive solver for this project because they retain the familiar physics-informed optimization framework of PINNs while replacing the rigid MLP-centric representation with adaptive Gaussian feature embeddings. Each Gaussian has a learnable center, scale, and feature weight. During training, Gaussians can migrate toward regions where the solution is more difficult to approximate, such as high-gradient or singular areas.

This is especially relevant for blade traps. In electrostatics, the potential itself may remain smooth in the vacuum region, but the field magnitude and curvature can vary sharply near the electrode boundaries and especially near blade tips or narrow openings. A solver that can reallocate representation capacity spatially is therefore highly desirable.

3.2.2. Field Representation

Let $\mathbf{x} \in \mathbb{R}^d$ denote a coordinate. We represent the solution through a Gaussian feature extractor:

$$\mathrm{FE}_\phi(\mathbf{x}) = \sum_{i=1}^{N} f_i \, G_i(\mathbf{x}), \tag{7}$$

where

$$G_i(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i)\right), \tag{8}$$

with learnable Gaussian centers $\boldsymbol{\mu}_i$, covariance parameters $\boldsymbol{\Sigma}_i$, and feature vectors f_i.

A lightweight neural refinement head then predicts the potential:

$$\Phi_{\phi,\omega}(\mathbf{x}) = \mathrm{NN}_\omega(\mathrm{FE}_\phi(\mathbf{x})). \tag{9}$$

In practice, the network head can be extremely small compared to a classical PINN, because most of the representational burden is carried by the Gaussian basis layer.

3.2.3. Why PIGs Are Well Matched to Blade Traps

The decision to use PIGs is not arbitrary. Blade Paul traps create a geometry in which:

- relevant features are *localized*,
- some regions require much higher representational capacity than others,
- the field behavior near edges matters disproportionately for downstream trap metrics.

PIGs match this regime well:

1. Gaussian kernels are local, unlike a global MLP basis.
2. Their centers and scales are adaptive, so capacity can migrate toward difficult regions.
3. Gaussian functions are smooth and differentiable, which is ideal for computing PDE residuals and higher-order derivatives.
4. The resulting model remains physics-informed and mesh-free in the training sense, while still behaving like an adaptive parametric mesh.

3.2.4. Physics-Informed Loss for the PIG Solver

The field solver will be trained with a composite loss:

$$\mathcal{L}_{\text{PIG}} = \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}}$$

(10)

The PDE residual term is:

$$\mathcal{L}_{\text{PDE}} = \frac{1}{|\mathcal{X}_\Omega|} \sum_{\mathbf{x}\in\mathcal{X}_\Omega} \left|\nabla^2 \Phi_{\phi,\omega}(\mathbf{x})\right|^2, \tag{11}$$

where $\mathcal{X}_\Omega$ are sampled interior collocation points.

The boundary condition term is:

$$\mathcal{L}_{\text{BC}} = \frac{1}{|\mathcal{X}_{\partial\Omega}|} \sum_{\mathbf{x}\in\mathcal{X}_{\partial\Omega}} \left|\Phi_{\phi,\omega}(\mathbf{x}) - V(\mathbf{x})\right|^2. \tag{12}$$

Optional symmetry constraints may be added for trap configurations designed to preserve mirror or rotational structure. Regularization can also be used to control pathological covariance growth, redundant Gaussian overlap, or drift away from the region of interest.

3.2.5. Expected Advantages Over PIXEL-Like Parametric Grids

A natural alternative would be a fixed-grid parametric representation such as PIXEL-style models. Those approaches are promising, but for blade-trap optimization they remain less attractive than PIGs. The main reason is that fixed-grid methods place representational resources at predetermined collocation locations, whereas PIGs can move those resources adaptively. In the present application, this matters because field difficulty is not uniform across the domain. The most challenging structures are precisely the localized regions near edges, gaps, and tip singularities. A static grid wastes parameters in uninformative regions, while adaptive Gaussians can migrate to the locations where additional resolution is needed most.

3.2.6. Sampling Strategy

The success of the PIG solver will also depend on a robust sampling policy.

- uniform interior collocation points for global consistency,
- boundary oversampling on electrode surfaces,
- boundary oversampling near tricky geometric regions
- residual-driven adaptive resampling as training progresses.

This complements the adaptive Gaussian representation and should improve both accuracy and data efficiency.

3.3. Shape Optimization via Coordinate Projection Networks (CPNs)

3.3.1. Why Choose CPNs for Shape Optimization?

The shape optimization module must satisfy three requirements:

1. it must be differentiable,
2. it must support nontrivial geometric deformations,
3. it must remain compatible with PDE training on domains with discontinuous material or boundary descriptions.

Coordinate Projection Networks satisfy these requirements elegantly. Rather than defining the design through a material density field or a smoothed indicator function, a CPN learns a mapping

$$\mathbf{x} = T_\psi(\mathbf{z}), \tag{13}$$

from reference coordinates $\mathbf{z}$ to deformed coordinates $\mathbf{x}$.

3.3.2. CPN Formulation

Let Ω_z denote the reference computational domain and $\Omega_x = T_\psi(\Omega_z)$ the deformed physical domain. The shape network is:

$$T_\psi : \Omega_z \to \Omega_x. \tag{14}$$

If $\mathbf{z}$ is a point in the reference geometry, its deformed physical coordinate is:

$$\mathbf{x} = T_\psi(\mathbf{z}). \tag{15}$$

The Jacobian of the mapping is:

$$J_\psi(\mathbf{z}) = \det \nabla_{\mathbf{z}} T_\psi(\mathbf{z}). \tag{16}$$

To avoid folding and preserve feasible geometry, we enforce:

$$J_\psi(\mathbf{z}) > 0 \tag{17}$$

through penalty terms. This makes sure the transformation stays a diffeomorphism, making sure there is no change in topology of the shape, since CPN's can't handle them.

3.3.3. Shape Loss Terms

The shape network is trained jointly with the field solver through a combination of trap-quality and geometric regularization losses:

$$\mathcal{L}_{\text{shape}} = \lambda_{\text{jac}}\mathcal{L}_{\text{jac}} + \lambda_{\text{vol}}\mathcal{L}_{\text{vol}} \tag{18}$$

Each term plays a distinct role in ensuring that the learned deformation improves trap performance while remaining geometrically valid.

Jacobian Regularization $\mathcal{L}_{\text{jac}}$. This term penalizes invalid local deformations by constraining the determinant of the Jacobian of the shape map T_ψ:

$$\mathcal{L}_{\text{jac}} = \frac{1}{|\mathcal{X}_{\text{shape}}|} \sum_{\mathbf{z}\in\mathcal{X}_{\text{shape}}} \left[\text{ReLU}(J_{\min} - \det \nabla T_\psi(\mathbf{z})) + \text{ReLU}(\det \nabla T_\psi(\mathbf{z}) - J_{\max})\right]^2 . \tag{19}$$

It prevents folding, inversion, or collapse of the geometry.

Smoothness Loss $\mathcal{L}_{\text{smooth}}$. This term penalizes rapidly varying or oscillatory deformations:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{|\mathcal{X}_{\text{shape}}|} \sum_{\mathbf{z}\in\mathcal{X}_{\text{shape}}} \|\nabla^2 T_\psi(\mathbf{z})\|^2. \tag{20}$$

It encourages smooth, fabrication-compatible electrode shapes.

Together, these losses ensure that the shape optimization is driven by trap-quality improvement while remaining stable, physically meaningful, and geometrically realistic.

3.4. Coupling the Shape Module with the PIG Solver

The full optimization loop is:

1. start from a reference blade geometry,
2. deform the geometry using CPN ,
3. sample interior and boundary points in the deformed domain,
4. solve for the RF potential using PIGs,
5. compute pseudopotential and trap-quality objectives,
6. backpropagate through the entire pipeline to update both solver and shape parameters.

The total loss may be written as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PIG}} + \mathcal{L}_{\text{shape}} + \lambda_{\text{access}}\mathcal{L}_{\text{access}}, \tag{21}$$

where

Access Loss $\mathcal{L}_{\text{access}}$. This term encourages geometries that improve optical or physical access to the trapping region by penalizing obstruction or proximity of electrodes to the trap center. A generic formulation is:

$$\mathcal{L}_{\text{access}} = \frac{1}{|\mathcal{X}_{\text{ROI}}|} \sum_{\mathbf{x} \in \mathcal{X}_{\text{ROI}}} \mathcal{P}(\mathbf{x}), \tag{22}$$

where $\mathcal{P}(\mathbf{x})$ measures geometric obstruction or distance-based penalties relative to the electrodes.

4. Experimental Plan

4.1. Validation Strategy

Validation is performed at three levels:

- **Physics:** PDE residual and boundary-condition accuracy,
- **Numerical:** comparison with external solvers,
- **Design:** geometric plausibility and trap-quality improvement.

4.2. Failure Modes to Monitor

Key risks include:

- insufficient resolution in the PIG solver,
- instability or collapse of Gaussian parameters,
- invalid or folded geometry from the CPN,
- objective-driven but physically unrealistic shapes,